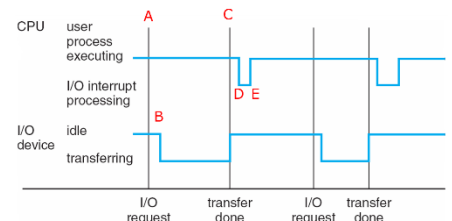


Midterm Exam 2022 (Operating System)

題目不回收。2~11為是非，12~16選擇題，每題都2分。17題之後請附解釋或計算過程，滿分:110，中括弧內的為單選的選項

1. (4%)選課班別為 ____ (A: 上午班, B: 下午班), 組別為 ____。
2. (3%)During the disk reading operation, disk controller is responsible(負責) for moving data [from controller's buffer to main memory. | **from disk to controller's buffer**]
3. (3%)Suppose that *gvar* is a global variable, then *gvar* is [dependent | **independent**(獨立) among sibling threads.
4. (3%)Which one is easier to port(移植) to different platform(平台)? [**API** | System Calls]
5. (3%)It is [**possible** | impossible] that user process executes instruction in kernel mode
6. (3%)In Multi-level Feedback Queue, system is in favor of [CPU-bound | **I/O bound**] process. 系統偏好哪一類行程?
7. (3%)**E**適用於互動系統的是哪一種排程?
(a) First-come, first-served, (b) Shortest job first, (c) Shortest remaining time first, (d) Priority, (e)Round robin
8. (6%)Divide by zero是[**同步** | 非同步]中斷, 進一步區分, 它是屬於[可修正的 | **不可修正**]
9. (3%)**BDE** Non-Preemptive system, 哪些事件會觸發CPU scheduler?
(a) Waiting-->Ready, (b) Running--> Ready, (c) Ready --> Running, (d)Running --> Waiting, (e)Running --> Terminated
10. (3%)**CD** (複選)Which of the following instructions should be privileged? (a)read from memory (b)issue system call (c)send command to Network Interface Card (d)modify PCB
11. (3%)**C**參考右圖, I/O裝置在哪一個時間發出中斷信號?
(a)A, (b)B, (c)C, (d)D, (e)E
12. (3%)Interrupt Controller是[**硬體** | 軟體]. 說出它的一個功能**可以讓多個裝置連接到CPU的中斷要求, 同時有多個裝置發出中斷時, 可以依據優先序循序處理**
13. (10%)There are four milestones in the developments of Operating System, (time sharing, batch, multi-processor, multi-programming).
(a) Please arrange them chronologically, (b) Since which milestone, it is capable of active-multiprogramming? 四個里程碑, 依照時間先後排列, **batch --> multi-programming --> time sharing --> multi-processor**哪個系統開始有 active-multiprogramming能力? **time sharing**
(a)batch (b)multiprogramming (c)time sharing (d)multi-processor
14. (10%)可否舉出一個實例說明, 即使只有一個單核CPU, 採取multi-thread 的方式會比較(相對於single thread)有效率(=執行時間較短)。如果認為不可能, 也請說明理由
15. (10%)碼頭的設施被颱風吹毀, 眾多的居民主動協助船員卸貨, 居民們排成一列人龍, 由岸邊的船一路延伸到倉庫, 貨物箱子就順著人龍進入倉庫. (a) 這種方式類似於[**任務平行化** | 資料平行化], (b) 如果採取另外一種平行化, 居民該如何做? (c)你會選擇哪一種平行化, 請說明理由。
16. (20%) The attributes of five processes are shown in the table below. (a) If 11 unit of CPU time was allocated(分配) to execute Process A, what are the possible states of Process A? (行程A的可能狀態有哪些?) **waiting or terminated**



(b)Schedule these processes by FCFS, draw the Gantt chart, and calculate the waiting time of each process. (c)same as before but using preemptive SJF, (d) same as before but using Round-Robin with time quantum=4. 畫出Gantt chart , 算出每一行程的等待時間。

Process	Arrival Time	Burst Time
A	0	11
B	2	7
C	5	28
D	8	16
E	13	2

waiting time = end time – arrival time – burst time = end time – (11, 9, 33, 24, 15)

(Time, Process)

FCFS:(0,A)-(11,B)-(18C)-(46D),(62E),(64null), end time=(11, 18, 46, 62, 64), waiting time=(0, 9, 13, 38, 49)

SJF: (0,A)-(11,B)-(18,E)-(20-D)-(36,C)-(64,null), end time=(11, 18, 64, 36, 20), waiting time = (0, 9, 31, 12, 5)

SRTF: (0,A)-(2,B)-(9,A)-(13,E)-(15,A)-(20,D)-(36,C)-(64,null), end time = (20, 9, 64, 36, 15), waiting time = (9, 0, 31, 12, 0)

RR(q=4): 會有兩組答案, 時間8的時候, B time-out, 而D進入系統, 兩個事件同時發生, B與D在誰先進入ready queue, 會有不同的結果。

B先進入, (Time, Process, Ready Queue):

(0,A11,null)-(4,B7,A7)-(8,A7,C28B3D16)-(12,C28,B3D16A3)-(16,B3,D16A3E2C24)-(19,D16,A3E2C24)-(23,A3,E2C24D12)-(26,E2,C24D12)-(28,C24,D12)-(32,D12,C20)-(36,C20,D8)-(40,D8,C16)

-(44,C16,D4)-(48,D4,C12)-(52,C12,null)-(64,null,null)

end time=(26, 19, 64, 52, 28), waiting time=(15, 10, 31, 28, 13)

D先進入:

<待完成>

17. (20%) Explain the following terms. Please demonstrate your knowledge in associate with those terms, instead of translations. Examples are very welcome. Clear description helps credit. (a)cycle stealing (b)context switch (c)degree of multiprogramming (d)device queue

解釋下列名詞。請展現對於作業系統的理解程度, 切勿翻譯了事。盡可能多用例子。字數太少, 資訊不足都很可能導致無法正確的評估。請盡量多寫, 寫清楚。